

Original Manuscript ID: Access-2026-38801

Original Article Title: “The Price of De-correlation: Quantifying the Agreement-Cost Trade-off in Heterogeneous Large Language Model Ensembles”

To: IEEE Access Editor

Re: Response to reviewers

Dear Editor,

Thank you for allowing a resubmission of our manuscript, with an opportunity to address the reviewers’ comments.

We are uploading (a) our point-by-point response to the comments (below) (response to reviewers, under “Author’s Response Files”), (b) an updated manuscript with yellow highlighting indicating changes (as “Highlighted PDF”), and (c) a clean updated manuscript without highlights (“Main Manuscript”).

The revision adds two substantial pieces of evidence. First, we now measure the correlation between agents’ errors directly rather than inferring it from label agreement, which is what the term “de-correlation” in our title asserts. Second, we ran a pre-registered replication of the entire protocol on a substantially different domain, 537 expert clinical-decision items, and report it whatever the outcome. Every number below is regenerated by the released code from the frozen data.

Best regards,

Samir Chincholikar
Robin Chawla

Reviewer #1, Concern #1

Reviewer comment:

Reduced agreement is not shown to improve accuracy or any decision-relevant outcome. Link the agreement-cost frontier to calibration, abstention, robustness, or tail-error reduction.

Author response:

- We agree that reduced agreement alone does not establish a downstream benefit.
- We added a selective-prediction analysis using mean agent conviction to rank ensemble decisions. AURC is 0.553 for HOM, 0.568 for HET-LITE, and 0.554 for HET. The HOM-HET difference is -0.001 with a 95% confidence interval of [-0.059, 0.060]. Under two-way equity and date resampling, the interval is [-0.113, 0.115]. Both intervals contain zero.
- Restricting evaluation to the most confident half of decisions does not materially reduce selective risk in any configuration. We therefore do not claim an improvement in calibration or abstention quality.
- We also evaluate a separate decision-relevant outcome, unanimous incorrect decisions. HOM produces unanimous incorrect directional decisions at 1.84 times the rate of HET. The equity-clustered 95% confidence interval is [1.42, 2.52], and the two-way interval is [1.13, 3.06]. Because this analysis is post hoc and no multiplicity adjustment is applied, we treat it as exploratory rather than confirmatory.

Author action:

- Section V-J, “Cross-Domain Replication: Clinical Knowledge,” subsection “Selective Prediction,” was added to report the risk-coverage analysis and uncertainty for the HOM-HET AURC contrast.
- Section V-H, “Exploratory Analysis of Consensus Errors,” was revised to report the unanimous-incorrect ratio with equity-clustered and two-way confidence intervals and to state explicitly that the analysis is exploratory.
- Section V-D, “The Agreement-Cost Frontier,” was revised to state that the frontier quantifies the cost associated with greater behavioral diversity and does not by itself establish improved downstream decision quality.

Reviewer #1, Concern #2

Reviewer comment:

The tier-matched control covers only two providers and no xAI equivalent. A fully tier-matched multi-provider design or model-level random effects is needed.

Author response:

- A fully three-provider tier-matched design remains unavailable because an equivalent efficient-tier xAI model was not available.
- We added an additional capability comparison in the clinical replication. Within the fully heterogeneous configuration, one same-provider pair and one cross-provider pair have nearly identical mean member accuracy, 0.951 and 0.955 respectively.
- Their error correlations are 0.954 and 0.607, a difference of 0.347 with a 95% confidence interval of [0.207, 0.507].
- We describe this as approximate capability matching. It reduces the capability confound but does

not eliminate the broader confounding between provider and other model-specific characteristics. We therefore do not identify provider as an independent causal factor.

Author action:

- Section V-J, “Cross-Domain Replication: Clinical Knowledge,” was revised to add the approximate capability-matched same-provider versus cross-provider comparison.
- Section V-K, “Robustness: Capability-Matched Control,” retains the pre-registered financial capability control, which preserves approximately 94% of the primary agreement contrast while holding model tier approximately constant.
- Section VII, “Threats to Validity,” subsection “Heterogeneity vs. capability confound,” was revised to state explicitly that provider remains confounded with model identity, training data, architecture, alignment, and effective capability.

Reviewer #1, Concern #3

Reviewer comment:

Findings come from one financial task, one prompt, three labels, and fixed five-agent ensembles. Test other domains, ensemble sizes, aggregation rules, and label structures.

Author response:

- We added a pre-registered replication on a substantially different clinical-knowledge task using 537 questions from the MMLU professional-medicine and clinical-knowledge subsets.
- The response structure changes from the three BUY, HOLD, and SELL labels used in finance to four answer options with one correct answer. The five-agent structure, three configurations, seeding procedure, temperature, number of runs, and estimators remain unchanged.
- The ordering reproduces. Agreement decreases by $\Delta\kappa = 0.072$ with a 95% confidence interval of [0.040, 0.104], compared with 0.336 in finance. Error correlation decreases by $\Delta\phi = 0.393$ with a 95% confidence interval of [0.325, 0.459], compared with 0.454 in finance.
- The agreement contrast therefore changes substantially more across the two tasks than the error-correlation contrast. We do not claim that the magnitude is invariant across domains.
- Ensemble size, aggregation rule, prompt format, and the particular model mixtures remain fixed and are stated explicitly as limitations.

Author action:

- Section V-J, “Cross-Domain Replication: Clinical Knowledge,” was added to report the pre-registered 537-item clinical replication.
- Section VII, “Threats to Validity,” subsection “Scope and generalizability,” was revised to distinguish the dimensions tested across domains from those that remain fixed.
- The clinical replication pre-registration and freeze receipt were added to the released reproducibility materials.

Reviewer #1, Concern #4

Reviewer comment:

The 1.84x unanimous-incorrect finding and HET-LITE recommendation are post hoc, lack CIs/p-values, and ignore multiplicity. Treat as hypothesis-generating only.

Author response:

- We now treat the consensus analysis explicitly as exploratory.
- The HOM-to-HET unanimous-incorrect ratio is 1.84. The equity-clustered 95% confidence interval is [1.42, 2.52]. Under two-way equity and date resampling, the interval is [1.13, 3.06]. Both exclude unity.
- The statistic and configuration comparison were selected after inspection of the data, multiple related comparisons were available, and no multiplicity adjustment is applied. The intervals are therefore not interpreted as confirmatory evidence.
- HET-LITE is no longer presented as a recommended ensemble design. Its observed cost efficiency is treated only as a hypothesis for prospective confirmatory evaluation.

Author action:

- Section V-H, “Exploratory Analysis of Consensus Errors,” was revised to report equity-clustered and two-way uncertainty for the unanimous-incorrect ratio.
- Section V-H was also revised to state the post hoc status of the analysis, note the absence of multiplicity adjustment, and remove the HET-LITE design recommendation.
- Section VI, “Discussion,” was revised so that the HET-LITE result is described as hypothesis-generating rather than deployment guidance.

Reviewer #1, Concern #5

Reviewer comment:

Seed support is best-effort, and the diagnostic is strongest only for HOM. Heterogeneous cross-run checks are confounded by repeated same-model draws. Add verification or sensitivity analysis.

Author response:

- We added a sensitivity analysis separating pairwise agreement according to whether the two agents use the same model.
- Same-model agreement is 0.800 in HOM, 0.794 in HET-LITE, and 0.862 in HET. Different-model agreement is 0.455 in HET-LITE and 0.490 in HET.
- Same-model agreement remains high across configurations, while different-model pairs exhibit substantially lower agreement. The reduction in ensemble agreement is therefore associated with the introduction of different-model pairs rather than a systematic decline in agreement among repeated instances of the same model.
- We also clarify that assigning distinct seeds removes the specific shared-seed artifact observed in the pilot but does not establish statistical independence of model responses.

Author action:

- Section V-E, “Shared-Seed Dependence and Measurement Validation,” subsection “Sensitivity: agreement by pair type,” was added to report the same-model and different-model pairwise agreement results.
- Section V-E, subsection “What unique seeds do and do not establish,” was added to state explicitly that

distinct per-agent seeds do not prove statistical independence.

- Figure 2 was revised to describe the change as assignment of a distinct reproducible seed to each agent and explicitly states that this does not establish statistical independence.
- Table 6 retains the within-run versus cross-run diagnostic and identifies the heterogeneous results as secondary consistency checks rather than direct independence tests.

Reviewer #1, Concern #6

Reviewer comment:

Table 4 and the text disagree on costs. Clarify per-decision cost, token accounting, and pricing assumptions.

Author response:

- We clarified the cost calculation. For each model call, metered prompt and completion tokens are priced using the per-million-token rates reported in the model registry.
- Costs are summed across the five agents within each ensemble decision and then averaged across cells within each configuration.
- The resulting costs are $\$6.48 \times 10^{-4}$ for HOM, $\$1.52 \times 10^{-3}$ for HET-LITE, and $\$2.84 \times 10^{-3}$ for HET. Moving from HOM to HET therefore increases inference cost by approximately $4.4\times$.
- Prices are fixed at those applicable during data collection. No discount, batching, or caching adjustment is applied. Failed requests returning no billable completion are excluded.

Author action:

- Section V-D, “The Agreement-Cost Frontier,” was revised to state the token accounting, aggregation, and pricing assumptions explicitly.
- Table 4 reports the corresponding per-ensemble-decision cost for each of the three configurations.

Reviewer #2, Concern #1

Reviewer comment:

Simplify index terms to 3 - 5 words only.

Author response:

- Done. The list has been reduced to five index terms.

Author action:

- The manuscript Index Terms were revised to: algorithmic monoculture, inter-rater agreement, large language models, model heterogeneity, and multi-agent systems.

Reviewer #2, Concern #2

Reviewer comment:

No clear explanation about fair comparison with current state-of-the-art approach.

Author response:

- This article does not propose a new ensemble algorithm, so a direct task-performance competition with existing ensemble methods is not the primary comparison.
- Existing methods such as mixture-of-agents, multi-agent debate, and diversity-based ensemble selection are generally evaluated through downstream performance. The present study instead measures how ensemble composition changes output agreement, error dependence, and inference cost under fixed task conditions.
- The relevant comparison is therefore methodological. The study separates output agreement from error correlation, controls the shared-seed artifact observed in repeated model sampling, quantifies statistical uncertainty, and incorporates inference cost.

Author action:

- Section V-M, “Relation to Existing Ensemble Methods,” was added to explain the basis of comparison with existing ensemble approaches.
- Section II-A, “LLM Ensembles and Mixture-of-Agents,” provides the corresponding literature context for mixture-of-agents, diversity-based selection, and heterogeneous ensemble approaches.

Reviewer #3, Concern #1**Reviewer comment:**

The present experiment confounds provider, model identity, capability, training data, architecture, and alignment. Add same-provider/different-model controls, more than one model per provider, balanced provider mixtures, and preferably a factorial analysis; otherwise rewrite all causal/provider-specific claims as configuration-specific observational findings.

Author response:

- We agree that provider is not experimentally separable from model identity, capability, training data, architecture, alignment procedure, and other model-specific characteristics in the primary design.
- We therefore interpret the provider comparisons as configuration-specific observational findings rather than isolated causal provider effects.
- The pre-registered financial capability control preserves approximately 94% of the primary agreement contrast when model tier is held approximately constant across OpenAI and Google.
- The clinical replication provides an additional approximate capability-matched comparison. One same-provider pair and one cross-provider pair have mean member accuracy of 0.951 and 0.955 respectively, while their error correlations are 0.954 and 0.607.
- These analyses narrow the capability confound but do not identify provider as an independent causal factor.

Author action:

- Section V-F, “Provider-Pair Agreement,” was rewritten as a configuration-specific observational analysis and now states explicitly that provider is bundled with other model characteristics.
- Section V-J, “Cross-Domain Replication: Clinical Knowledge,” was revised to include the approximate capability-matched same-provider versus cross-provider comparison.

- Section V-K, “Robustness: Capability-Matched Control,” retains the pre-registered financial capability control.
- Section VII, “Threats to Validity,” subsection “Heterogeneity vs. capability confound,” was revised to state explicitly that provider cannot be isolated causally in the present design.

Reviewer #3, Concern #2

Reviewer comment:

Recompute uncertainty while accounting for both repeated observations within equities and shared analysis dates/common market conditions. Justify the bootstrap unit formally, evaluate two-way equity/date clustering or an appropriate multiway resampling strategy, and regenerate all main and exploratory confidence intervals accordingly.

Author response:

- Equity is the pre-registered primary resampling unit because the repeated observations in this design are nested within securities: each equity is evaluated on 12 analysis dates, and every response for that equity shares company-specific context. Analysis date is the second, crossed source of dependence, because all securities evaluated on a given date share common market conditions. Neither margin nests inside the other, so they are resampled independently rather than hierarchically.
- The two-way procedure draws equities with replacement and dates with replacement, independently, retains the sampled equity-date intersections, and preserves multiplicity on both margins: a cluster drawn twice contributes twice. It uses 2,000 draws at the pre-registered seed and recomputes the full estimator inside every draw.
- Primary HOM-HET agreement contrast, $\Delta\kappa = 0.3363$: [0.3035, 0.3689] under equity clustering (100 clusters), [0.2989, 0.3656] under date clustering (12 clusters), [0.2786, 0.3877] under two-way resampling (1,200 intersections). All exclude zero.
- The two pre-registered secondary agreement contrasts are measured on the same grid and are now reported under the same scheme. HOM-HET-LITE $\Delta\kappa = 0.249$: equity [0.218, 0.280], two-way [0.208, 0.290]. HET-LITE-HET $\Delta\kappa = 0.087$: equity [0.067, 0.105], two-way [0.045, 0.123]. Both continue to exclude zero.
- The capability-matched control is measured on the same 100 equities and 12 dates and asserts that its interval excludes zero, so it receives the same treatment. $\Delta\kappa(\text{HOM-SameTier}) = 0.315$: equity [0.288, 0.341], two-way [0.270, 0.361]. Both exclude zero.
- Every exploratory contrast that carries an inferential claim is also reported both ways. Error correlation $\Delta\phi = 0.454$: equity [0.376, 0.534], two-way [0.328, 0.586]. The common-subset specification counting HOLD as an error, $\Delta\phi = 0.309$: equity [0.262, 0.354], two-way [0.227, 0.387]. Unanimous-incorrect ratio 1.84: equity [1.42, 2.52], two-way [1.13, 3.06]. Pile-on $\Delta = 0.189$: equity [0.160, 0.216], two-way [0.140, 0.232]. AURC $\Delta = -0.001$: equity [-0.059, 0.060], two-way [-0.113, 0.115]. No conclusion changes under the wider scheme: the intervals that excluded the null still exclude it, and the AURC interval still contains zero.
- Two groups of intervals are deliberately not given a two-way treatment, for reasons of design rather than convenience. The temperature-sensitivity analysis is a small supporting subgrid of eight equities and two analysis dates; two date clusters cannot support date-clustered or two-way inference, so its intervals remain clustered on equities only and the manuscript now states that this analysis contributes no independent multiway evidence. The clinical replication is indexed by MMLU items rather than by

equities and dates, so the financial equity-date bootstrap does not apply to it; its intervals are clustered on items.

- The Wilson intervals in the secondary directional-accuracy table are marginal and descriptive. They are retained as descriptive only and are not the basis of any inferential claim, so replacing them with a clustered interval would imply an inference the manuscript does not make.
- One correction follows from this work rather than from a change of method. The implementation used for the robustness table and for the unanimous-incorrect ratio collapsed each draw to a set of equities, so an equity drawn twice counted once and each draw used only the distinct equities of a 100-draw multiset. That is subsampling, not a cluster bootstrap, and it understated the variance. Multiplicity is now preserved, which is why the equity-clustered robustness row reproduces the primary interval exactly and why the unanimous-incorrect interval is [1.42, 2.52] rather than the narrower [1.51, 2.35] previously reported. The resampling code now asserts multiplicity on both margins and fails if it is ever lost, so the error cannot recur silently.

Author action:

- Section III-D, “Uncertainty Quantification,” justifies equity as the primary cluster, identifies date as the crossed dimension, and defines the date-clustered and two-way procedures.
- Table 5 reports the primary contrast under equity, date, and two-way resampling.
- Section V-A, “Heterogeneity Reduces Ensemble Agreement (Primary),” now reports the two-way intervals for both pre-registered secondary contrasts alongside Table 2.
- Section V-K, “Robustness: Capability-Matched Control,” now reports the two-way interval for the capability-matched contrast.
- Section V-C, “Agreement, Error Correlation, and Decorrelation,” reports two-way intervals for both error-correlation specifications.
- Section V-H, “Exploratory Analysis of Consensus Errors,” reports two-way intervals for the unanimous-incorrect ratio and the pile-on contrast.
- Section V-I, “Selective Prediction,” reports the two-way AURC interval.
- Section V-L, “Robustness: Temperature Sensitivity,” states that the subgrid spans two analysis dates, which is too few clusters for date-clustered or two-way inference, and that the analysis therefore contributes no independent multiway evidence.

Reviewer #3, Concern #3

Reviewer comment:

Fleiss’ kappa measures agreement, not correlation. Either remove “de-correlation”/“less-correlated judgments” throughout the title, abstract, figures, discussion, and conclusion, or add actual dependence/error-correlation analyses that justify those terms. Explicitly distinguish output agreement, error correlation, sampling independence, model diversity, and provider diversity.

Author response:

- We took the second option and now measure error dependence directly.
- For every pair of agents within an ensemble cell, we compute ϕ between binary error indicators. Error correlation decreases from 0.984 for HOM to 0.742 for HET-LITE and 0.530 for HET. The HOM-HET contrast is 0.454 with a 95% confidence interval of [0.376, 0.534]. Under two-way resampling, the

interval is [0.328, 0.586].

- Because treating HOLD as an abstention changes the evaluated subset, we repeat the analysis with HOLD counted as an error. The corresponding HOM-HET contrast is 0.309 with a 95% confidence interval of [0.262, 0.354]. The two-way interval is [0.227, 0.387].
- Mean per-agent error rates remain similar across configurations, so the decline in ϕ reflects lower error dependence rather than differences in average correctness.
- The manuscript now distinguishes output agreement, error correlation, sampling independence, model diversity, and provider diversity. Decorrelation is defined specifically as a reduction in error correlation measured by ϕ , while κ measures categorical agreement.

Author action:

- Section V-C, “Agreement, Error Correlation, and Decorrelation,” was added to define the five concepts and report the ϕ analysis under both treatments of HOLD.
- Section V-A was retitled “Heterogeneity Reduces Ensemble Agreement (Primary)” so that the κ result is described explicitly as an agreement result.
- The Abstract was revised to report the ϕ contrast alongside the κ contrast and to distinguish reduced error dependence from reduced agreement.
- Section VI, “Discussion,” and Section VIII, “Conclusion,” were revised so that decorrelation claims refer to error dependence rather than categorical agreement. The Conclusion now states that model diversity can reduce both output agreement and error correlation.

Reviewer #3, Concern #4

Reviewer comment:

HOLD-induced abstention creates different evaluation subsets across configurations, so the reported hit rates are not directly comparable. Perform a proper selective-prediction/coverage analysis, use paired and clustered inference, provide uncertainty for unanimous-error and pile-on statistics, and stop using overlapping marginal Wilson intervals as the main inferential evidence.

Author response:

- We added a selective-prediction analysis using mean agent conviction to rank decisions. The HOM-HET AURC difference is -0.001 with a 95% equity-clustered confidence interval of [-0.059, 0.060]. The corresponding two-way interval is [-0.113, 0.115]. Both contain zero.
- The unanimous-incorrect ratio is 1.84 with an equity-clustered interval of [1.42, 2.52] and a two-way interval of [1.13, 3.06].
- For pile-on, the HOM-HET difference is 0.189 with an equity-clustered interval of [0.160, 0.216] and a two-way interval of [0.140, 0.232].
- We also report error correlation with HOLD counted as an error, which fixes the evaluated subset across configurations. The resulting HOM-HET ϕ contrast is 0.309 with a 95% confidence interval of [0.262, 0.354] and a two-way interval of [0.227, 0.387].
- Wilson intervals are retained only in the descriptive secondary accuracy analysis and are no longer the principal inferential evidence.

Author action:

- Section V-I, “Selective Prediction,” was added to report risk-coverage analysis and paired clustered AURC inference.
- Section V-H, “Exploratory Analysis of Consensus Errors,” was revised to report equity-clustered and two-way uncertainty for unanimous incorrect decisions and pile-on.
- Section V-C, “Agreement, Error Correlation, and Decorrelation,” was revised to report the common-subset error-correlation analysis with HOLD counted as an error.
- Section V-G, “Secondary Directional-Accuracy Analysis,” retains the directional hit-rate analysis as secondary and does not use overlapping marginal Wilson intervals as the principal inferential evidence.

Reviewer #3, Concern #5

Reviewer comment:

Clarify that unique seeds do not prove statistical independence; provide complete API parameters and versions, exact prompts/system messages, seed-support evidence, retry records, total attempted versus successful API calls, exact preregistration timestamp evidence, and archived immutable code/data. Correct Figure 3 so that the shared-seed pilot, independent-seed pilot, and full-study quantities compare the same estimand.

Author response:

- The manuscript now states explicitly that assigning distinct per-agent seeds removes the demonstrated shared-seed artifact but does not establish statistical independence.
- The confirmatory grid contains 54,000 planned responses, all of which ultimately returned valid outputs. Completing the grid required 94,566 API requests. Of the 40,566 unsuccessful requests, 40,544 were rate-limit or quota responses and 22 were transport or schema failures.
- Every request attempt is retained in the released log. The reproducibility materials contain request settings, complete prompts, prompt hashes, model identifiers, seeds, prices, experimental configuration, pre-registration records, archived responses, and the frozen analysis materials.
- Figure 3 now contains only two like-for-like pilot quantities. Both measure the HOM-HET agreement contrast on the same pilot grid and model set. The contrast is 0.485 under a shared seed and 0.113 under distinct per-agent seeds.
- The full-study within-minus-cross-run diagnostic is reported separately in Table 6 and is no longer combined with the Figure 3 estimand.

Author action:

- Section III-C, “Independent-Draw Estimand,” was revised to document the per-agent seed construction and the limits of seed-based reproducibility.
- Section IV-C, “Models,” records the model identifiers, provider assignments, inference settings, timestamps, seeds, prompt hashes, token usage, costs, and archived responses.
- Section IV-D, “Pre-Registration, Freezing, and Reproducibility,” documents the frozen design, pre-registration hash, immutable dataset, and deterministic analysis pipeline.
- Section IV-E, “Collection Provenance and Retry Accounting,” was added to report the 94,566 attempted requests, 54,000 valid outputs, failure breakdown, and retained attempt history.
- Section V-E, “Shared-Seed Dependence and Measurement Validation,” was revised to distinguish removal of the shared-seed artifact from proof of statistical independence.

- Figure 2 was revised to show the distinct per-agent seed construction and explicitly state that this does not establish statistical independence.
- Figure 3 was reduced to the two directly comparable pilot HOM-HET agreement contrasts.
- Table 6 reports the full-study within-run versus cross-run diagnostic separately.

Reviewer #3, Concern #6

Reviewer comment:

Replicate the main experiment on at least one substantially different domain/task and preferably multiple prompt formats, ensemble sizes, and model mixtures. The present single BUY/HOLD/SELL financial task-whose accuracy is itself near chance and whose label marginals vary strongly by configuration-is insufficient to establish a general principle about heterogeneous LLM ensembles.

Author response:

- We added a pre-registered replication using 537 questions from the MMLU professional-medicine and clinical-knowledge subsets.
- The task has four response options with one correct answer rather than the three BUY, HOLD, and SELL labels used in the financial experiment. Chance performance is 25%, and the evaluated models perform substantially above that level.
- Choice order is shuffled deterministically within each item to remove positional imbalance in the published answer key.
- The main ordering reproduces. Agreement decreases by $\Delta\kappa = 0.072$ with a 95% confidence interval of [0.040, 0.104], while error correlation decreases by $\Delta\phi = 0.393$ with a 95% confidence interval of [0.325, 0.459].
- The agreement contrast is substantially smaller than in finance, while the error-correlation contrast attenuates much less. We therefore report the cross-domain replication without claiming that the magnitude is invariant across tasks.
- Prompt format, five-agent ensemble size, majority-vote aggregation, and the tested model mixtures remain fixed and are stated explicitly as limitations.

Author action:

- Section V-J, “Cross-Domain Replication: Clinical Knowledge,” was added to report the pre-registered 537-item clinical replication.
- Section VII, “Threats to Validity,” subsection “Scope and generalizability,” was revised to state that the ordering has been evaluated across two substantially different tasks while prompt format, ensemble size, aggregation rule, and model mixtures remain fixed.
- The clinical replication pre-registration and freeze receipt were added to the released reproducibility materials.